General homework guidelines

- Problem sets are due on Friday by 10:00 a.m., uploaded to Gradescope. Late assignments will not be accepted, so make sure to budget enough time for scanning and uploading.
- All turned in work must be legible. If the grader cannot read your answer, the grader will not be able to give you points for your answer. Your submissions should be neat and clear, with the pages correctly selected on Gradescope. You may lose points for anything that makes your solution difficult for the grader to read.
- All answers should be presented as clearly as possible. Even if an answer is correct, it may lose points if it is overly complicated, poorly structured, or otherwise hard to follow. This includes using full English sentences and proper grammar and punctuation.
- Include the text of the question you are answering. If you are asked to prove a statement, use a Theorem/Proof format.
- For the first homework assignment, on Problem 1 tell me either your favorite outdoor activity. This way I'll know that you read this document.

Proof writing guidelines

- Start your proof with "**Proof.**" The body of a proof should generally start by stating the assumptions you are making and then stating what you are trying to prove.
- You make it clear when a proof ends. A sentence like "This completes the proof" is good. It's standard practice to then use a symbol like □ at the bottom right corner of a completed proof. You can use any symbol you like.
- If you introduce a variable, explain what it means. For example, "Let x be a real number such that $x > 2$" is great. It helps to give specific objects names for example, "Let A be the set of integers larger than 5."
- The textbooks listed on the syllabus generally provide good examples of proof writing. Your proofs don't have to be identical in style, but they probably shouldn't be wildly different.
- If you use the phrase "by definition," say which definition you're referring to.
- Avoid using words like "clearly," "obviously," etc. They do not help your proofs, and are often a way that people hide flaws in logic.

Advice on LaTeX

- A great skill to learn is using LaTeX to type up your mathematics. This is not required but highly encouraged, particularly if you want to pursuit a math major.
- If this is your first experience, I would recommend using **Overleaf.com** as your LaTeX editor.
- This **WikiBook** is a nice introduction for writing math with LaTeX.
- I will provide a `.tex` template file for each homework assignment, see the "Schedule and lecture notes" module in the course Canvas.
- I will be happy to answer any LaTeX-related questions. If there is enough interest I can also make a short video tutorial, write me an e-mail if that's the case.

Grading scale

The TA will grade all of the problems on each assignment. ln a hypothetical 1–5 scale, each question will receive credit (approximately) as follows.

Points	Characteristics
5	Problem is completely, correctly, and clearly answered.
4	Very minor deficiencies in one of the above categories.
3	Substantial progress has been made, but noticeable flaws.
2	Some correct elements or outline, but largely incorrect.
1	Only a small amount of "scratch work" is provided, no general solution.
0	You only wrote down the problem statement.